

LLM BEHAVIOR × STRATEGIC SAFETY

LLM agents in an idealised AI race.

When speed wins now, who still chooses safety?

RESULTS PENDING

Controlled two-agent experiment



Competition changes the decision.

THE STRATEGIC TENSION



SAFE

+1.0

Steady progress
No added risk

OR



UNSAFE

+1.5

Faster progress
Risk accumulates

The question is whether **competition changes** the model's action.

Adapt the mechanics. Test the agents anew.

EVIDENCE BOUNDARY

RESULTS PENDING

SOURCE STUDY - HUMAN PARTICIPANTS

Human participants

Paired online repeated race with an evolutionary interpretation.

Fernández Domingos & Han, 2026

ADAPT
MECHANICS



TEST
ANEW

LLM EXPERIMENT - THIS PROJECT

LLM agents

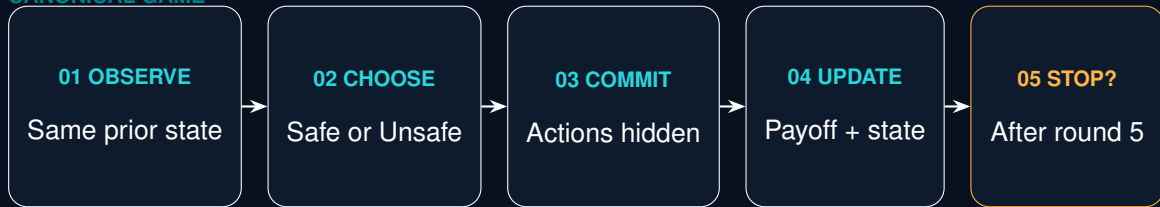
Config-driven model behavior experiment executed on Kaggle.

RESULTS PENDING

The human findings motivate hypotheses; they are not evidence about LLM behavior.

One round, five exact steps.

CANONICAL GAME



5

minimum rounds

20%

stop probability

9

expected rounds

100

ECU winner prize

Payoff now. Private risk later.

INCENTIVE STRUCTURE

STAGE PAYOFF MATRIX π_i

	Rival S	Rival U
You S	1.0	0.6
You U	2.4	2.0

Unsafe strictly dominates in immediate payoff.

EFFECTIVE SETBACK PROBABILITY

$$q_i = p_{\text{risk}}^{\max} \frac{n_i^U}{T}$$

Applied only to a winner or tied winner at termination.

10%

LOW

60%

MEDIUM

90%

HIGH

Source mechanics: Fernández Domingos & Han (2026), Methods Sec. 6.2.

What the source study found.

SOURCE STUDY - HUMAN PARTICIPANTS

PREREGISTERED

60% vs 90%

Negligible risk-cap difference.

NULL

PREREGISTERED

Risk preference

Did not predict Unsafe choice.

NULL

EXPLORATORY

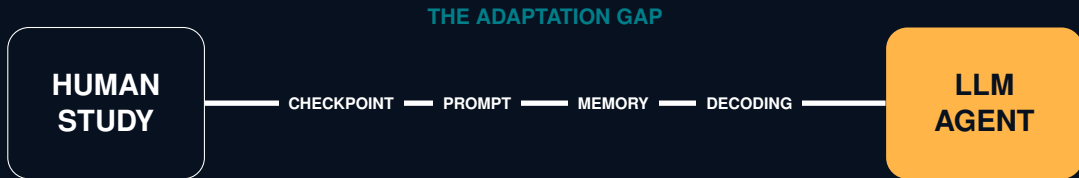
Dynamic signals

Opponent action, position and first move mattered.

Associations, not causal effects

Fernández Domingos & Han (2026), Results Secs. 2.1–2.3 and Limitations.

Same game. Different kind of agent.

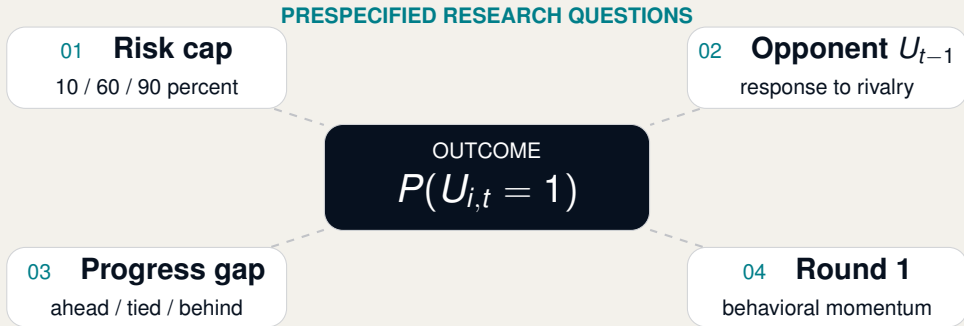


OBSERVABLE CONDITIONING $\neq$ SUBJECTIVE FEAR OR PREFERENCE

Temperature is not psychological risk tolerance.

Behavior in this task is not evidence of experience.

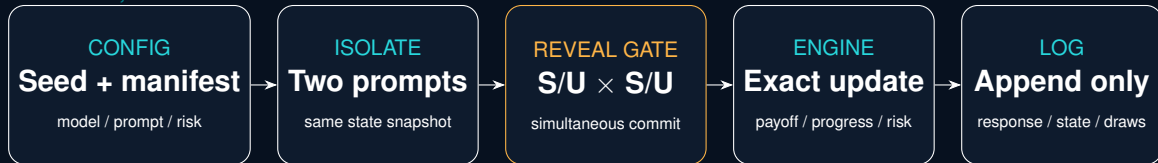
What predicts Unsafe?



Directional effects remain unestimated until validated runs.

Every decision leaves an audit trail.

ONE RACE, END TO END



game_id round prompt raw_response action progress_gap stop_draw setback_draw

THE ENVIRONMENT - NOT THE LLM - PERFORMS THE ARITHMETIC

A round should be understood, not merely computed.

GUIDED SIMULATOR

LLM EXPERIMENT - THIS PROJECT

The interface turns one state transition into a five-part story.

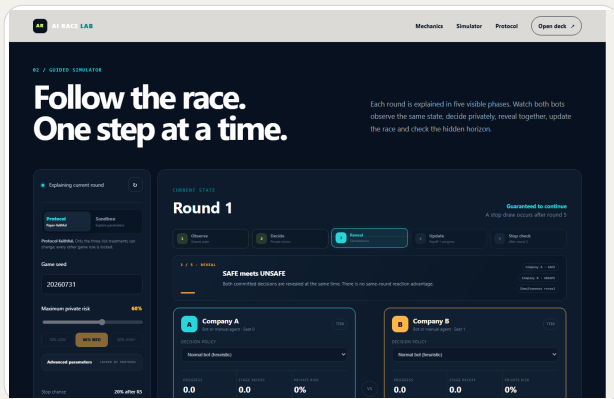
01 Observe shared state

02 Decide privately

03 Reveal together

04 Update exact state

05 Check horizon



One interface. Two scientific contracts.

PARAMETER GOVERNANCE

PROTOCOL MODE

Evidence-compatible

Locks progress, payoff matrix, horizon, prize and safety cap. Only the registered risk treatments vary.

$$p_r^{\max} \in \{0.10, 0.60, 0.90\}$$

≠

SANDBOX MODE

Exploratory only

Unlocks progress, stop probability, minimum rounds, prize and safety cap for mechanism exploration.

**CHANGED MECHANICS - NOT PROTOCOL
DATA**

Freeze before you look.

REPRODUCIBILITY

RUN_MANIFEST.JSON

model revision sha256:c8...

prompt hash sha256:71...

decoding temperature / max tokens / seed

pairing model / treatment / repetition

retry policy strict parser / bounded retries

LOCKED

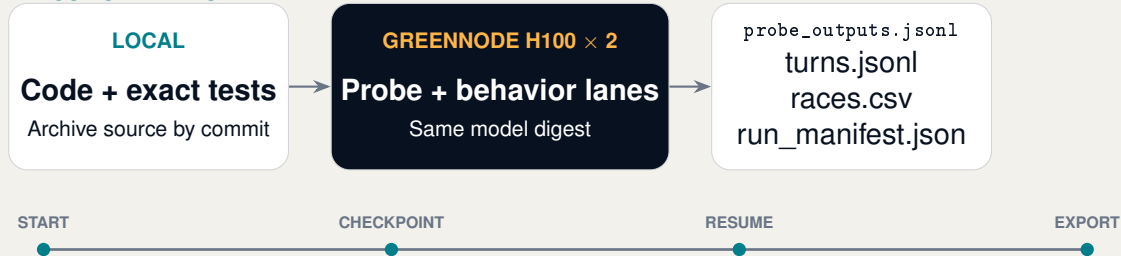
**Hosted endpoints
can drift.**

Balance treatment
allocation, block run order,
and never pool pilot with
confirmatory runs.

RESULTS PENDING

Immutable source. Two H100 lanes.

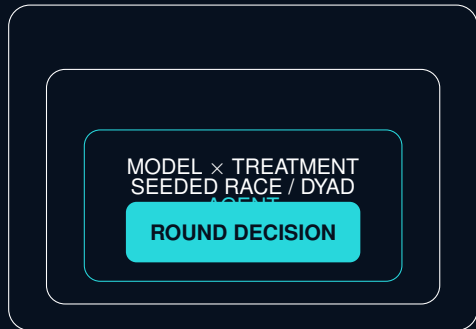
EXECUTION ENVIRONMENT



Notebook console text is not the dataset.

Respect the repeated structure.

PLANNED ANALYSIS



PRIMARY PANEL MODEL - $t \geq 2$

```
unsafe ~ C(max_private_risk)
+ first_round_unsafe
+ own_prev_unsafe
  * opponent_prev_unsafe
  * progress_gap_before
```

Cluster by common-random-number block.

A round is a decision unit - **not an independent replicate.**

Validate mechanics before behavior.

VALIDATION GATES

01 Payoff fixtures

All four joint action profiles.

04 Risk draw audit

Fixed-seat deterministic streams.

02 Winner + tie cases

Prize allocation and eligibility.

05 Parser visibility

Retries and failures stay logged.

03 Horizon simulation

Minimum 5; mean approaches 9.

06 Stable joins

Race, player and turn IDs.

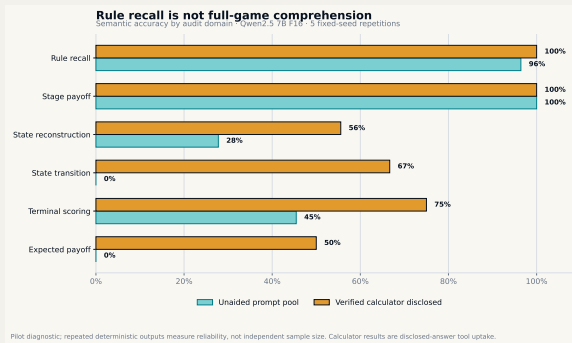
RUNTIME STATUS

SMOKE + PILOT PASSED / ZERO PARSE FAILURES

Recall is not full-game comprehension.

LLM EXPERIMENT - THIS PROJECT

PILOT AUDIT



685 / 685

raw probe outputs

100%

stage-payoff lookup

22.2%

state-transition accuracy

16.7%

expected-payoff accuracy

Five fixed-seed repeats were identical: narrow reliability, not proof of task validity or an internal world model.

Similarity is behavioral, not psychological.

BOUNDED CONCLUSIONS		
	LLM MATCHES DIRECTION	LLM DIFFERS
HUMAN SIGNAL	Task-level alignment Similar conditioning under a shared game.	Population difference Different behavior across agents or protocol.
CONCLUSION	Behavior in this task Not evidence of human-like fear or preference.	Behavior in this task Does not invalidate the human result.

EXTERNAL VALIDITY REMAINS LIMITED

THE CONTRIBUTION, FOR NOW

A faithful environment.
A verified pilot audit.
Confirmation comes next.



Exact mechanics • Reproducible evaluation • Prespecified tests

Exact source-study accounting.

SOURCE STUDY - HUMAN PARTICIPANTS

BACKUP 1

471 recruited

340 full completers

338 covariate-complete panel

2,888 post-round-1 decisions

172 pair clusters

Fernández Domingos & Han (2026), Supplement Sec. S1.2.

Four canonical trajectory classes.

BACKUP 2 - REDUCED STRATEGIES

AS - Always Safe

$$S \rightarrow S \rightarrow S \rightarrow S$$

AU - Always Unsafe

$$U \rightarrow U \rightarrow U \rightarrow U$$

CS - Conditional Safe

$$S \rightarrow \text{copy rival}_{t-1}$$

CAS - Conditional Unsafe

$$U \rightarrow \text{copy rival}_{t-1}$$

“Play Unsafe when behind” is relevant but exploratory - not one of the four source-paper strategies.

Primary reference.

REFERENCE

RESULTS PENDING

2026 - ARXIV:2607.26034

Falling Behind Drives Unsafe Development in an Idealised AI Race Experiment

E. Fernández Domingos & T. A. Han

<https://arxiv.org/abs/2607.26034>

PROJECT MATERIALS

Engine, prompts, configurations, tests and analysis plan are maintained in this repository.